

PQC Migration Prize Competition

Hosted by the Japanese Government (METI/NEDO) | BGIN: coordination & evaluation

Blockchain Governance Initiative Network (BGIN)

Multi-year initiative: 2026–2028

Community briefing — July 2026

The initiative

- A multi-year, internationally open **prize competition** on **post-quantum (PQC) migration** for public blockchains.
- **Hosted by the Japanese Government (METI/NEDO).**
- **BGIN** coordinates internationally and runs a **neutral evaluation testbed** (BSafe.network).
- Deploy **NIST-selected / standardized PQC** — **not** invent new PQC primitives.

🔗 **Takeaway:** Competition / evaluation → BGIN Standard → ISO → Japan crypto-asset security criteria.

Why it matters — standards path

- The prize is a catalyst; the lasting goal is **shared practice**.
- Outputs are meant to become a **BGIN Standard**.
- Then advance toward an **ISO Standard** (via ISO/TC 307; BGIN is Category A liaison).
- And feed **Japan's crypto-asset security criteria** for operators and market infrastructure.

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What it is — and what it is not

It is

- A push to **accelerate and de-risk** migration to quantum-resistant signatures and advanced key management.
- **Transparent, reproducible** evaluation on a geographically distributed neutral testbed.
- Focused on **deployment** on live / public networks.
- Globally open to applicants (funded by Japan).
- A pipeline toward **standards and security criteria** — not a one-off prize ceremony.

🔗 **Takeaway:** The hard problem on public chains is deployment, evaluation, testbeds, and migration governance — not selecting new PQC primitives.

It is not

- **Not inventing new PQC primitives** — starts from NIST process outputs (e.g. ML-DSA and related primitives).
- **Not fear-based timelines** — threat narratives are scenarios, not a single “Q-day” calendar claim.
- **Not chain endorsement** — ecosystem-neutral forum.

Why it matters — path to standards

- 1 **BGIN Standard** — codify migration evidence, rubrics, and evaluation practice accepted in the multi-stakeholder forum.
- 2 **ISO Standard** — carry mature BGIN work toward **ISO/TC 307** (BGIN holds Category A liaison).
- 3 **Japan's crypto-asset security criteria** — feed domestic security criteria / guidance so operators and market infrastructure can reference shared practice.

Competition / evaluation → BGIN Standard → ISO → Japan criteria

- **Prize ≠ end state:** awards incentivize deployable results; **standardization is the lasting benefit.**
- International participants help shape criteria that later inform **both global ISO work and Japanese domestic security baselines.**
- Neutral testbed evidence is designed to be **citable** in standards and supervisory conversations — not vendor marketing.

🔗 **Takeaway:** Engage early if you want migration metrics that can travel from competition → BGIN → ISO → Japan criteria.

Roles — Japanese Government, BGIN, and NIST

Lane	Role
Japanese Government (METI/NEDO)	Host and organizer — publishes the call, sets rules, funds and awards the prize; later channel for reflecting practice into Japan's crypto-asset security criteria .
BGIN	International coordination and neutral evaluation testbed ; multi-stakeholder dialogue; produces candidate BGIN Standards with a path to ISO/TC 307 (Category A liaison).

NIST (not a competition lane)

Algorithm / parameter authority for the PQC primitives used in the competition. Measurement and evaluation-methodology cooperation is sought where appropriate.

 **Takeaway:** Host funds the prize; BGIN turns shared evidence into standards-ready artifacts; NIST remains the reference for which PQC primitives are in scope.

Evaluation focus

- › Comparative evaluation is anchored on **Bitcoin** and **Ethereum** migration evidence.
- › These two networks provide concrete, comparable targets for on-chain deployment, performance, and governance fit.
- › **General-purpose / other-chain work is not precluded:** chain-agnostic technologies from any ecosystem are welcome when they can be mapped to Bitcoin and/or Ethereum evidence.

🔗 **Takeaway:** Focus the scoring lens on Bitcoin and Ethereum; do not shut the door on general-purpose approaches that transfer to those targets.

Cross-cutting themes

- › **PQC migration:** formats, aggregation, L2/rollups, hybrid / staged migration, throughput & decentralization cost.
- › **Advanced key management:** MPC, multisig, ZKP; t -of- n availability; recovery; custody boundaries.
- › **Governance & incentives:** stakeholder fit, dormant / vulnerable assets (discussion input — not a policy decree).

NIST-style dual track

- (a) Theoretical security review.
- (b) Implementation / performance on a neutral, geographically distributed testbed (**BSafe.network** concept; order of ~10–15 nodes; multi-region; multi-month runs planned).

🔗 **Takeaway:** Open criteria and reproducible results are the raw material for a **BGIN Standard**, ISO submission, and Japan security-criteria uptake.

Rubric directions (indicative)

- Migration evidence and reproducibility.
- Verifier cost; latency; block / mempool overhead.
- Compatibility / rollback / governance fit.
- Wallet / UX and L2 impact where relevant.

Neutrality and conflict of interest

- **Mandatory COI disclosure** for evaluators and node operators.
- **Role separation:** submitters (and close affiliates) do not judge that round; test-node operators running an artifact under evaluation are recused (or a documented mitigation applies).
- **Sponsor vs. judgment:** prize funding (Japanese Government / METI/NEDO) and coordination (BGIN) must not compromise technical neutrality.
- BGIN is **not** the algorithm authority and does **not** endorse a vendor or chain.

🔗 **Takeaway:** Neutrality is what makes outputs usable as **standards inputs** (BGIN → ISO → Japan criteria), not as a private scorecard.

Who can participate

Entrants (indicative)

- › Bitcoin and Ethereum developers / project teams (primary evaluation focus).
- › Other-chain / general-purpose teams — **not precluded**, when results map to Bitcoin and/or Ethereum evidence.
- › Wallet and infrastructure providers.
- › Academia and industry R&D groups.

🔗 **Takeaway:** Participants help write the practice that can later sit in BGIN / ISO / Japan security documents — not only compete for a prize.

Evaluation node hosts (indicative)

- › Accredited universities.
- › Nonprofit organizations.
- › Neutral R&D labs.

Eligibility details (sanctions, vendor roles, regulatory risk) will be finalized in the official call.

Indicative schedule — 2026 into early 2027

Period	Owner	Activity
2026 May–Jul	BGIN	Lecture series; community alignment.
2026 Sep 1–3	BGIN (GDC)	PQC session(s) for policy alignment.
2026 Sep ~17	Academic (CBT)	PQC-related academic session.
2026 Oct 9–12	Ethereum Fdn.	Quantum Retreat (Cambridge) — specs & testbeds.
2026 Oct 15–16	BGIN (Block 15)	PQC / evaluation session (NIST-facing, DC).
2026 Oct 19–23	CSS / BWS	Academic drafting of guidelines & criteria.
2026 Dec 13	SSR WS	Co-located PQC workshop (Baltimore); formal call text targeted around this window.
2027 Jan	METI/NEDO	Competition applications open (planned).

All dates provisional until confirmed by organizers. Early work already aims at standards-ready vocabulary and rubrics.

Workshop (not the main SSR PC)

- **Date (working):** Sunday 13 December 2026, ~10:00–16:00, Baltimore (ssresearch26.umbc.edu).
- Co-located workshop parallel to the protocol-analysis workshop — **not** a parallel track of the main conference.
- BGIN carries operational / financial responsibility (coordination with SSR OC).
- Programme: submitted talks + invited / panel (archival full papers may use SSR 2027 pathways as agreed).

🔗 **Takeaway:** SSR is an academic checkpoint on the same path that later supports BGIN Standard → ISO → Japan criteria.

Working milestones

- **Program Committee locked:** by 31 July 2026.
- **CfP:** early–mid August 2026.
- **Paper due: October 2026** (exact date set with the PC).
- Venue for discussing migration, measurement, and evaluation criteria ahead of the formal prize call.

Indicative schedule — 2027 to 2028

Period	Owner	Activity
2027 Feb	Academic / BGIN	Workshops (e.g. CoDecFin; Block discussions).
2027 Jun	Academic (ICBC)	Competition-related workshop.
2027 Sep	METI/NEDO	Application deadline (planned).
2027 Oct	BGIN / CSS	Screening / presentation formats.
2027 Nov–	METI/NEDO / BGIN	Evaluation rounds begin.
2027 Dec–2028 mid	Testbed	Performance evaluation phase (indicative).
2028 mid–Sep	METI/NEDO	Winner selection & prize (planned).
2028 Oct–	BGIN / Academic	Toward BGIN Standard; ISO/TC 307 & Japan criteria follow-on.

Prize award is a milestone; **standardization follow-through** is a primary objective.

Ways to engage — menu

Channel	What it means (indicative)
Eval. committee	Join as a member of the competition evaluation committee: help draft call text, metrics, and rubrics in Autumn 2026 (Phase A); optional later evaluation rounds (Phase B) under the host process.
Technical advisor	Track-specific input (e.g. Bitcoin / Ethereum / measurement) without winner-selection votes.
SSR workshop PC / speak	Shape the 13 Dec workshop programme; review around the October due date; or contribute as speaker / panelist.
Testbed node	University / neutral lab hosts a BSafe.network evaluation node (contracting later; role separated from judging).
Submit / build	Enter the prize when the official call opens (planned 2027).
Discuss & amplify	Discourse, sessions (GDC, Block, CSS/BWS), mutual visibility with complementary industry roadmaps.

 **Takeaway:** Every channel above can contribute to the same end-state: **BGIN Standard** → **ISO**



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From prize to standards

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BGIN Standard → ISO → Japan crypto-asset security criteria. | July 2026 briefing.



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